

SQL Operators

MySQL (SQL/PL-SQL) · Unit I · Lecture 5

BCA · SEMESTER PROGRAM

PRESENTED BY: RENUKA PARMAR



Quick Recap

What We've Covered So Far

Database & DBMS

Concept of data storage, DBMS vs RDBMS

SQL & MySQL

Structured Query Language and MySQL engine

SQL Categories

DDL, DML, DCL, TCL, DQL

SQL Data Types

INT, VARCHAR, DATE, FLOAT, CHAR, etc.



Today we build on this foundation — SQL Operators bring queries to life.



Lecture 5 · Learning Objectives

By the End of This Lecture, You Will Be Able To...



Define SQL Operators

Understand what operators are and why they are needed in SQL queries.



Classify All Operator Types

Identify Arithmetic, Relational, Logical, and Special operators.



Apply Operators in Queries

Write correct SQL using WHERE, HAVING, and JOIN clauses.



Avoid Common Mistakes

Recognize and fix typical operator errors in student-level SQL.



Foundation Concept

What Is an Operator?

An **operator** is a symbol or keyword that tells SQL to perform a specific operation on one or more values — producing a result.

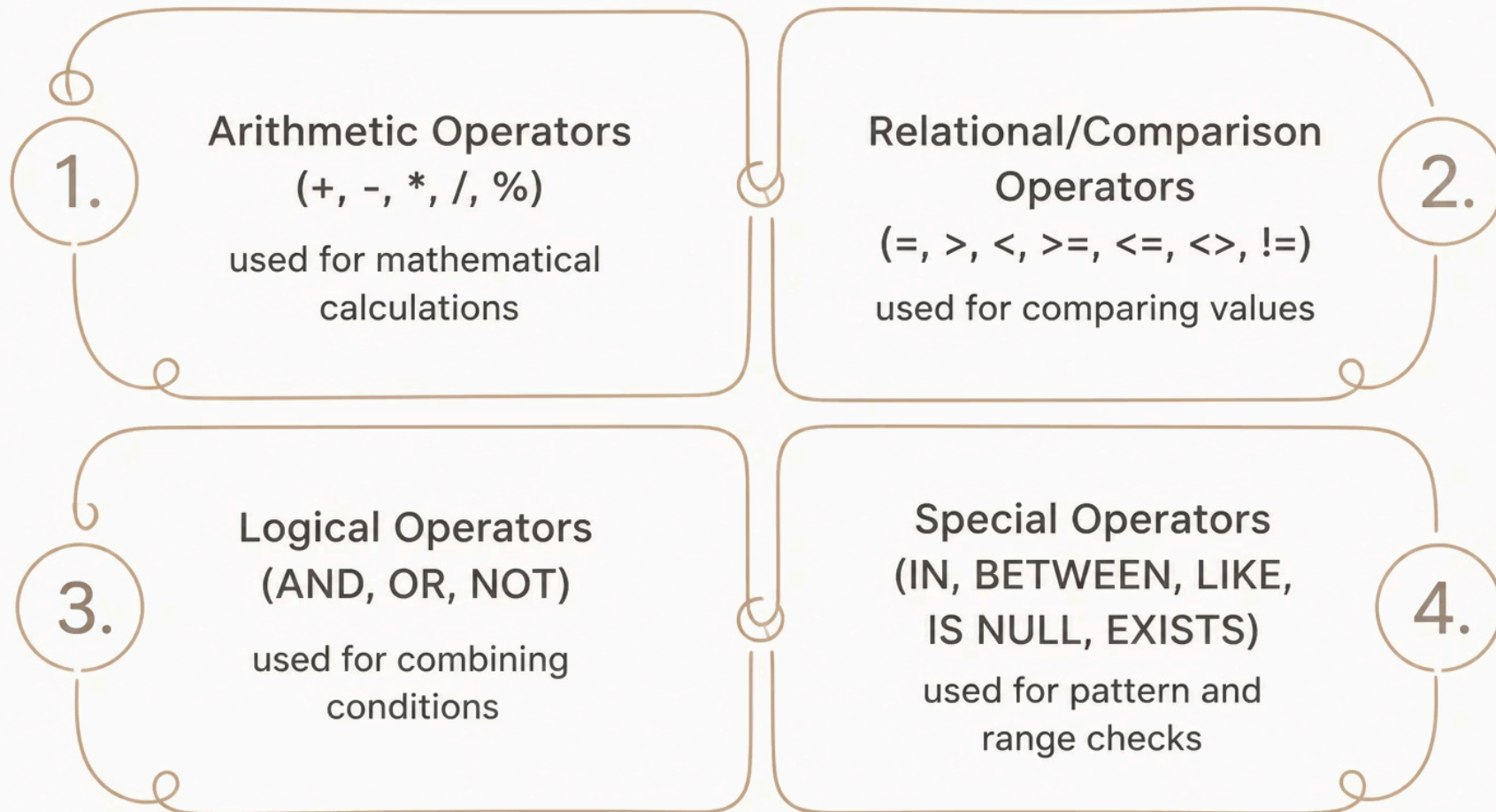
Why Do We Need Operators?

- To **filter** rows using conditions (e.g., salary > 50000)
- To **compute** values on the fly (e.g., price * quantity)
- To **combine** multiple conditions (e.g., AND, OR)
- To **search patterns** and check NULL values

Overview

Classification of SQL Operators

Each category serves a distinct purpose in building powerful, precise SQL queries.



Category 1

Arithmetic Operators

Operator	Meaning	Example
+	Addition	salary + bonus
-	Subtraction	price - discount
*	Multiplication	qty * rate
/	Division	total / count
%	Modulus (Remainder)	marks % 10

Real-World Example

Calculate **net salary** for all employees:

```
SELECT emp_name,  
       salary + bonus AS net_pay  
FROM employee;
```

- ❑ Arithmetic operators work inside SELECT and WHERE clauses on numeric columns.

Category 2

Relational (Comparison) Operators

=

= (Equal)

dept = 'CS'

>

> / <

marks > 75

>

>= / <=

age <= 25

≠

<> / !=

city != 'Delhi'

Purpose

Compare two values — returns **TRUE** or **FALSE**.
Used primarily in WHERE clauses to filter rows.

Example

```
SELECT * FROM student  
WHERE marks >= 60;
```

Category 3

Logical Operators

AND

Both conditions must be TRUE.

```
WHERE age > 18 AND city  
= 'Pune'
```

OR

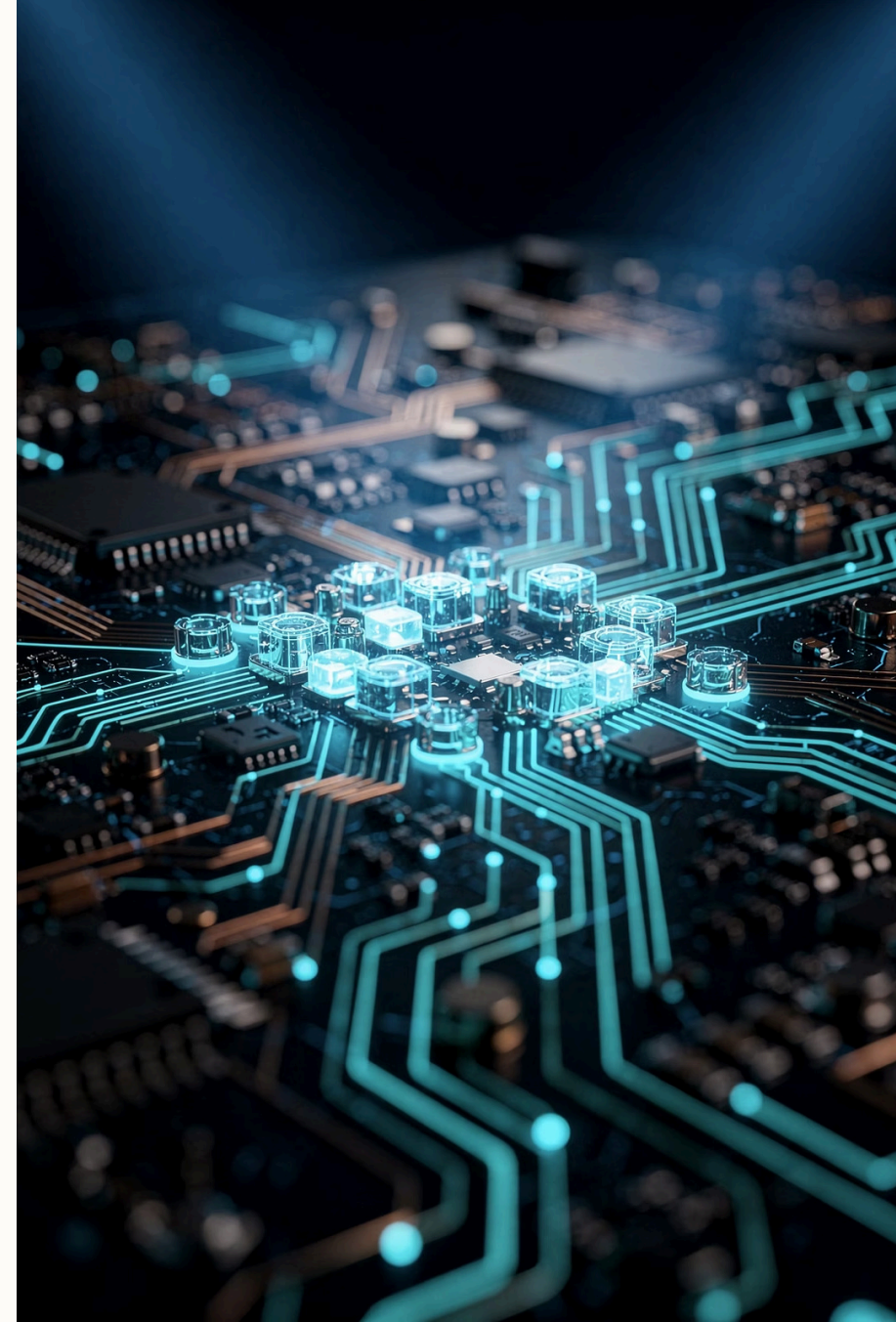
At least one condition must be TRUE.

```
WHERE dept = 'CS' OR  
dept = 'IT'
```

NOT

Reverses/negates the condition result.

```
WHERE NOT city = 'Delhi'
```



Category 4

Special Operators

IN

Checks if a value exists in a list.

```
WHERE city IN ('Mumbai', 'Pune',  
'Delhi')
```

BETWEEN

Checks if a value is within a range (inclusive).

```
WHERE marks BETWEEN 60 AND 90
```

LIKE

Matches a pattern using % or _ wildcards.

```
WHERE name LIKE 'R%'
```

IS NULL

Tests whether a column has no value stored.

```
WHERE email IS NULL
```

EXISTS

Checks if a subquery returns any row.

```
WHERE EXISTS (SELECT * FROM  
orders ...)
```

Comparison

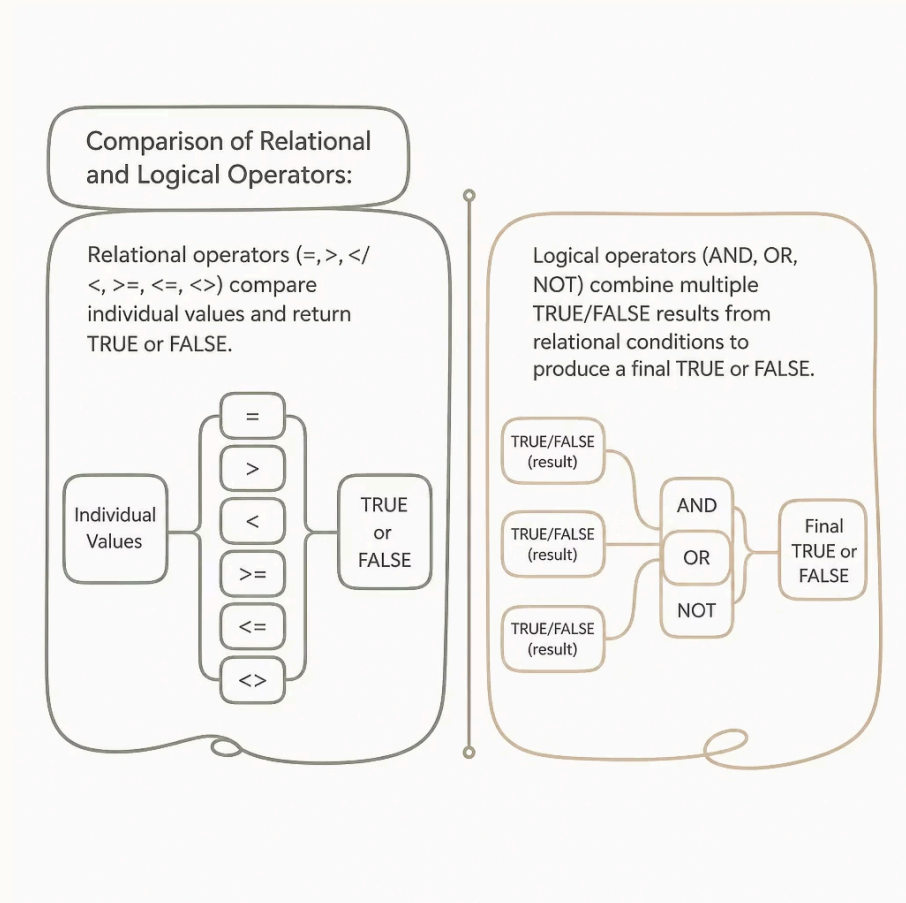
Arithmetic vs. Relational Operators

Aspect	Arithmetic	Relational
Purpose	Perform calculations on values	Compare two values
Returns	A numeric result	TRUE or FALSE
Operators	+ - * / %	= > < >= <= <>
Used In	SELECT clause (computed columns)	WHERE clause (row filtering)
Example	salary * 1.1 AS hike	WHERE salary > 50000

❏ Both can appear together: WHERE (salary + bonus) > 60000

Comparison

Relational vs. Logical Operators



Key Insight

Relational operators evaluate a single condition.
Logical operators combine multiple conditions.

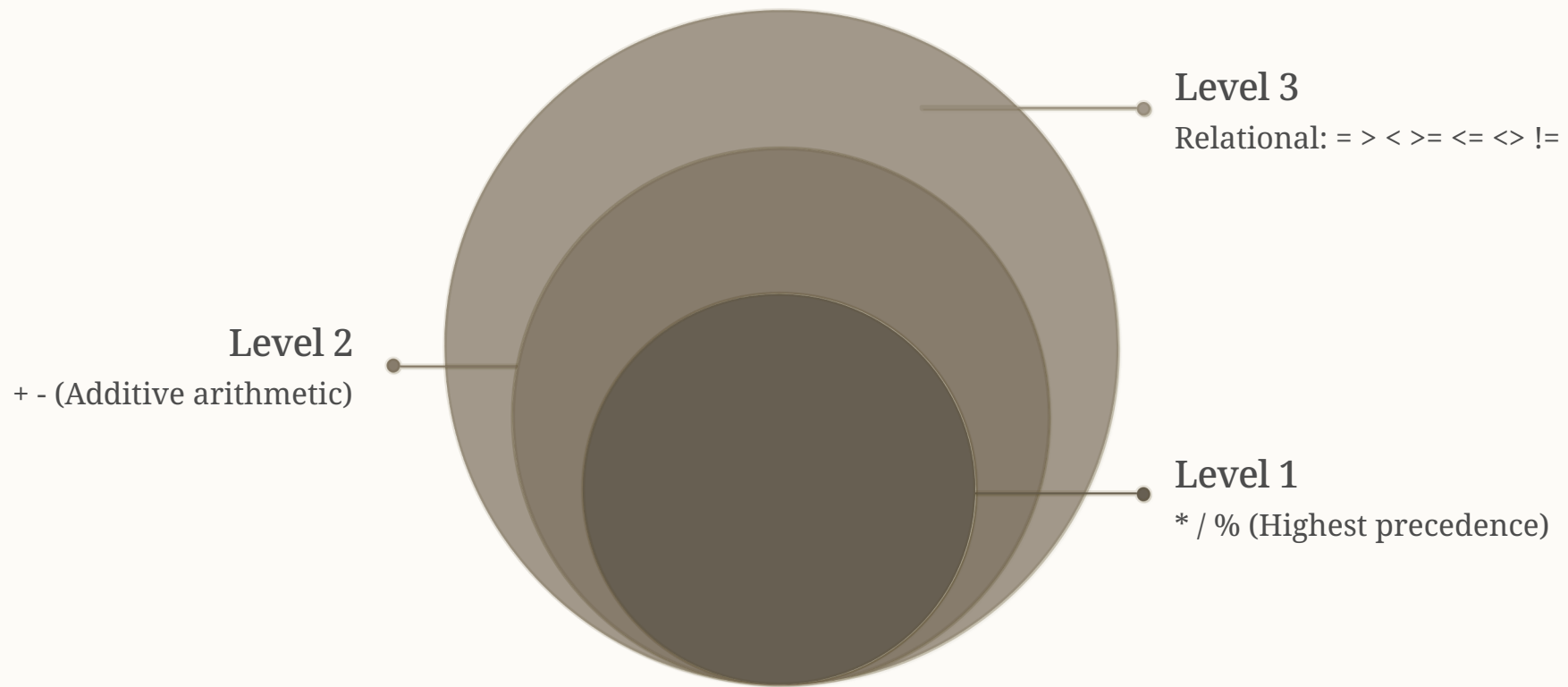
```
WHERE marks > 60  
AND dept = 'BCA';
```

Here $>$ and $=$ are relational; AND is logical — they work **together**.

Important Concept

Operator Precedence in SQL

When multiple operators appear in a single expression, SQL follows a strict **order of evaluation**:



Use **parentheses ()** to override default precedence and make your intent explicit — this prevents logic errors in complex WHERE conditions.

Hands-On Examples

Real-World SQL Using Student & Employee Tables

1

Filter High Scorers

```
SELECT name FROM student  
WHERE marks >= 75 AND dept = 'BCA';
```

2

Salary Range Check

```
SELECT emp_name FROM employee  
WHERE salary BETWEEN 30000 AND 70000;
```

3

Name Pattern Search

```
SELECT * FROM student  
WHERE name LIKE 'A%';
```

4

Multiple Departments

```
SELECT * FROM employee  
WHERE dept IN ('HR','Finance','IT');
```

Usage Context

Where SQL Operators Are Used



Key Clause Examples

- **WHERE:** WHERE age > 20
- **HAVING:** HAVING COUNT(*) > 5
- **SELECT:** salary * 1.1 AS new_sal
- **JOIN ON:** ON s.dept_id = d.id
- **UPDATE:** WHERE emp_id = 101



Watch Out!

Common Mistakes While Using SQL Operators

→ Using = for NULL

Wrong: `WHERE email = NULL` — Always use `IS NULL` or `IS NOT NULL`.

→ BETWEEN Range Order

Lower value must come first: `BETWEEN 10 AND 50`, not `BETWEEN 50 AND 10`.

→ LIKE Without Wildcards

`LIKE 'Raj'` behaves like `= 'Raj'`. Use `%` or `_` for pattern matching.

→ Ignoring Operator Precedence

Always use parentheses: `WHERE (A OR B) AND C` to avoid unexpected results.

Apply Your Knowledge

Practical Scenarios — Choose the Correct Operator

Scenario 1

Find students whose names start with the letter "S".

→ Use: **LIKE 'S%'**

Scenario 2

Find employees earning between ₹25,000 and ₹60,000.

→ Use: **BETWEEN**

Scenario 3

Find all students NOT from the 'CS' department.

→ Use: **NOT or <>**

Scenario 4

Find employees whose phone number is missing.

→ Use: **IS NULL**

Classroom Activity

Match the Requirement with the Correct Operator

#	Requirement	Correct Operator
1	Students with marks more than 80	>
2	Employees in Mumbai, Pune, or Nagpur	IN
3	Students whose email is not filled	IS NULL
4	Products costing between ₹100 and ₹500	BETWEEN
5	Name starts with 'M' and ends with 'a'	LIKE 'M%a'
6	Marks above 50 AND attendance above 75%	AND

☐ Discuss each row with your neighbor — explain WHY that operator fits before revealing the answer.

Summary Table — All SQL Operators

Category	Operators	Purpose	Example
Arithmetic	+ - * / %	Mathematical calculations	salary * 1.1
Relational	= > < >= <= <>	Compare values, return T/F	marks >= 60
Logical	AND OR NOT	Combine conditions	age>18 AND city='Pune'
Special – Range	BETWEEN	Inclusive range check	sal BETWEEN 30K–70K
Special – List	IN	Match any value in a list	dept IN ('CS','IT')
Special – Pattern	LIKE	Match string patterns	name LIKE 'A%'
Special – Null	IS NULL	Check for missing values	email IS NULL

Assessment

Quiz + Assignment

5 MCQs — Pick the Correct Answer

- Which operator checks if a value is within a range? **(a) IN (b) BETWEEN (c) LIKE (d) EXISTS**
- Which is used to match patterns? **(a) = (b) LIKE (c) IN (d) >=**
- What does IS NULL check? **(a) Zero (b) Empty string (c) Missing value (d) False**
- Operator precedence: which is evaluated first? **(a) AND (b) OR (c) NOT (d) =**
- WHERE dept IN ('CS','IT') is equivalent to? **(a) BETWEEN (b) OR (c) AND (d) LIKE**

Home Assignment

Using the **student** table, write SQL queries for:

1. Students with marks between 60 and 90
2. Students from 'BCA' OR 'MCA'
3. Students whose email IS NULL
4. Names starting with 'P' using LIKE

Submit in next class as a handwritten query sheet.

Key Takeaways



Operators Are the Engine of SQL

Every filter, calculation, and condition in SQL depends on operators.



Four Categories, One Goal

Arithmetic, Relational, Logical, and Special operators each solve different data-retrieval problems.



Precedence Matters

Misplaced operators can silently return wrong data. Always use parentheses in complex conditions.



July
17

Next Lecture (6): SQL Expressions — computed columns, aliases, CASE expressions, and string functions.

